

Expectation-Maximization: Application to Gaussian Mixture Model Parameter Estimation

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Outline

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Application to
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- 1 Mixture Models
- 2 Maximum Likelihood Estimation
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1 Mixture Models

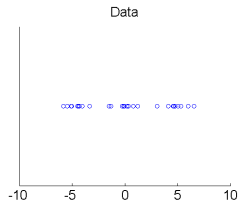
2 Maximum Likelihood Estimation

3 Expectation-Maximization

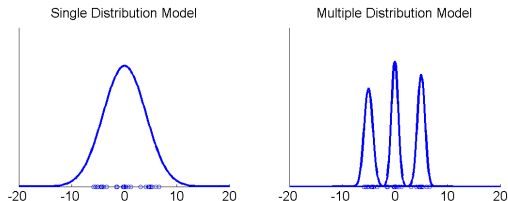
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Mixture Models - Motivation

We want to construct a model for this data:



Which is better?



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Mixture Models - Introduction

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What is a mixture model?

- Fancy word for a weighed sum of distributions
- Also known as “mixture distribution”
- Weights are called “mixing coefficients”
- $p(x) = \pi_1 p_1(x) + \pi_2 p_2(x) + \dots + \pi_k p_k(x)$
- Used when one analytic distribution does not accurately model the situation

Gaussian Mixture Models (GMMs)

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Informally...

- Sum of weighted Gaussians

Formally...

$$p(x) = \sum_{k=1}^K \pi_k N_{\mu_k, \sigma_k}(x)$$

- K : the number of distributions in the mixture
- π_k : the weight of the k^{th} distribution
- μ_k : the mean of the k^{th} distribution
- σ : the standard deviation of the k^{th} distribution

Notational Convenience

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- We will deal with only 1D distributions in the analysis

k^{th} 1D Gaussian Distribution

$$N_{\mu_k, \sigma_k}(x) = N_k(x)$$

Weighted Distribution

$$W_k(x) = \pi_k N_k(x)$$

GMM Example

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- $\mu_1 = -2, \sigma_1 = 2, \pi_1 = .3$
- $\mu_2 = 1, \sigma_2 = 1, \pi_2 = .5$
- $\mu_3 = 3, \sigma_3 = .5, \pi_3 = .2$

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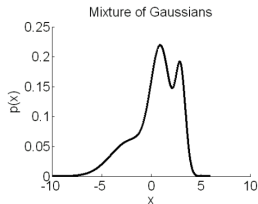
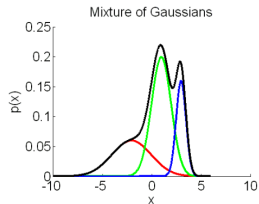
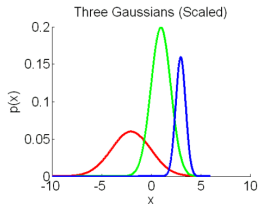
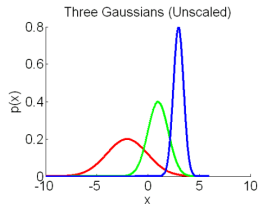
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GMM Details

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Applications

- Mixing coefficients must sum to 1 so that the resulting distribution has the properties of a probability distribution

$$\sum_{k=1}^K \pi_k = 1$$

- Each distribution must be a proper distribution

$$\int_{-\infty}^{\infty} p_k(x) dx = 1, \quad 0 \leq p_k(x_i) \leq 1 \forall i$$

- Mixing coefficients can be interpreted as priors
 - If we have not yet observed a sample and we must guess which distribution it came from, it is more likely to have come from the distribution with the largest weight (a la Salmon/Seabass problem)

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Maximum Likelihood Estimation (MLE)

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Note/Disclaimer:

MLE is an extremely important topic that many whole books have been written about. This is the short version.

- Find the maximum of the joint distribution of all of the observations over the parameter space

$$MLE = \arg \max_{\theta} p(x_1, x_2, \dots, x_n | \theta) = \arg \max_{\theta} \prod_{i=1}^N p(x_i | \theta)$$

- θ is a vector of parameters of the distribution
- This is called “Parametric Density Estimation” (vs. Non-Parametric ie. Parzen Windows)

MLE Simplification

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Applications

- We can simplify the problem by taking the log of the function
- Log function is monotonic, so the $\max(\text{original function}) = \max(\log \text{ of function})$
- The properties of logs give us the huge advantage of converting the product into a sum

$$MLE = \arg \max_{\theta} \sum_{i=1}^N \log(p(x_i))$$

MLE Simplification for Gaussians

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Applications

- In general, this optimization must be performed numerically
- However, in the case of Gaussians, since the function is entirely an exponential function (more formally, it is in the “exponential family of distributions”), when we take the log, it is tremendously simplified
- Equation of a Gaussian

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

MLE Simplification for Gaussians

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$$\begin{aligned}MLE &= \arg \max_{\theta} \prod_{i=1}^N \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \\&= \arg \max_{\theta} \sum_{i=1}^N \log \left[\frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \right] \\&= \arg \max_{\theta} \sum_{i=1}^N \log \left(\frac{1}{\sigma \sqrt{2\pi}} \right) + \log \left[\exp \left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \right] \\&= \arg \max_{\theta} \sum_{i=1}^N \log \left(\frac{1}{\sigma \sqrt{2\pi}} \right) - \frac{(x_i - \mu)^2}{2\sigma^2} =\end{aligned}$$

MLE For Gaussians

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Gaussian Distribution: 2 Parameters

$$\theta = \begin{pmatrix} \mu \\ \sigma \end{pmatrix}$$

MLE of μ , $\hat{\mu}$

Taking the derivative of the likelihood function with respect to μ and maximizing (ie. setting the result to 0 and solving for μ), we find the MLE of the mean is the average of the values of the observations

$$\hat{\mu} = \frac{1}{N} \sum_{n=1}^N x_n$$

MLE For Gaussians

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MLE of $\sigma, \hat{\sigma}$

Differentiating with respect to σ^2 , we find the MLE of the variance to be

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \hat{\mu})^2$$

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Applications

What it is not...

- Finding the expected value of a maximization function
- Maximizing an expectation function

What it is...

- A two step process for finding MLE solutions
 - 1 Expectation
 - 2 Maximization

But ML is so easy... why do we need EM?

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Applications

- Not usually, only for Gaussians
- Generally it is not only hard, but impossible
- What if an entire term didn't go away when we differentiate?
- What if this didn't happen?

$$\log[\exp(\dots)]$$

Responsibilities

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Applications

What you will see in text books...

- Define $z \in \{0, 1\}^k$
- $$\gamma_k(z_k) = \frac{p(z_k = 1)p(x|z_k = 1)}{\sum_{j=1}^K p(z_j = 1)p(x|z_j = 1)}$$
- Clearly,
$$\gamma_k(z_k) = \frac{\pi_k \mathbf{N}(x|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathbf{N}(x|\mu_j, \Sigma_j)}$$

Responsibilities

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What they were trying to say...

- The “responsibility” the k^{th} distribution takes for “explaining” the observation is the value of the k^{th} function evaluated at x divided by the sum of all of the weighted distributions evaluated at x

$$\gamma_k = \frac{W_k(x)}{\sum_{j=1}^K W_j(x)}$$

Responsibilities Diagram

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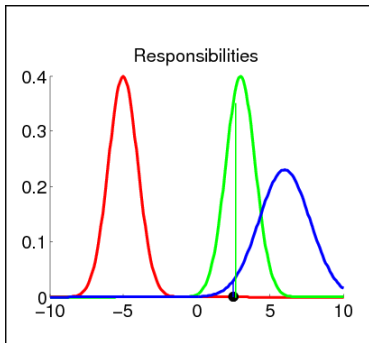
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$$\gamma_k = \frac{W_k(x)}{\sum_{j=1}^K W_j(x)}$$

$$\gamma_{red} = \frac{0}{0 + .05 + .35} = 0$$

$$\gamma_{blue} = \frac{.05}{0 + .05 + .35} = .125$$

$$\gamma_{gree} = \frac{.4}{0 + .05 + .35} = .875$$

E-Step, M-Step

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E-Step (Expectation)

Use the current values of the parameters to evaluate the responsibilities

M-Step (Maximization)

Re-estimate the distribution parameters (using the current responsibilities) to maximize the likelihood of the data

Increase in Likelihood Function

- Guaranteed!
- However, may not find a global maximum

K-Means Clustering

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Problem Statement

Given a set of data, create the best k groups (clusters)

Algorithm

- Initialize: Pick either random points from the data or random points in the space as the cluster centers
- E-Step: Assign each point the label of the closest cluster center
- M-Step: Find the centers of the current clusters
- Iterate

“Hard” assignment

- Where generally EM is “soft” assignment

K-Means Demo

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GMM Example

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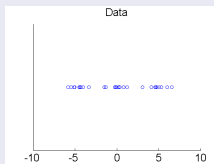
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Problem Statement

Given data that is known to come from a GMM:

- Estimate the parameters of the mixture distributions
- Determine which distribution each sample came from
- Ex. We have this data, and we know that it came from a mixture of 3 Gaussians:



GMM Example (Cont.)

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Algorithm

- EM!
- Does both of these tasks simultaneously

GMM EM Example: Initialization

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Applications

Means and variances

- Many choices...
- Choose random, reasonable values
- Choose means which cover the space
- Set variances to equal, reasonable values

Responsibilities

- K-Means clustering - higher weight to cluster with more points (remember, we can interpret the mixture weights as priors)
- $\gamma_k = \frac{1}{K}$

GMM EM Example: E-Step

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Applications

Evaluate responsibilities

- There are Nk responsibilities, the responsibility that the k^{th} distribution takes for generating the n^{th} data point

$$\gamma_{nk} = \frac{W_k(x_n)}{\sum_{j=1}^K W_j(x_n)}$$

GMM EM Example: M-Step

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Means

- $\hat{\mu}_k = \frac{1}{N} \sum_{n=1}^N x_n$ becomes $\hat{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} x_n$
- N_k is the *effective* number of points assigned to the k^{th} distribution, ie. $N_k = \sum_{n=1}^N \gamma_{nk}$
- New mixing coefficients are $\pi_k = \frac{N_k}{N}$

GMM EM Example - M-Step Variances

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Same idea...

$$\hat{\sigma}^2 = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} (x_n - \hat{\mu})^2$$

GMM Demo

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Conclusions

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What did we learn?

- What is a mixture model?
- Basics of maximum likelihood estimation
- How to use EM to iteratively solve for the MLE solution